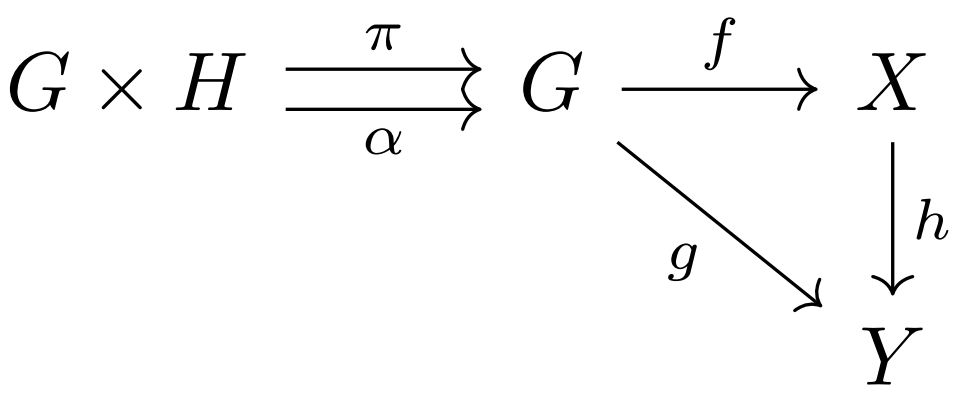
$G \times H$
π
α
G
f
X
g
h
Y